

Research

Realistic synthetic dataset generation for cyber-physical systems: a performance evaluation

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Abstract

Cyber-physical systems (CPS) represent the seamless integration of physical and digital systems connected through network of sensors, actuators, and controllers. Research in the field of CPS is crucial given its application in healthcare, smart cities, transportation, energy management and autonomous vehicles. The development and validation of models, algorithms, and control systems for CPS heavily relies on the availability of quality CPS datasets. However, conducting data collection experiments on complex CPS are constrained by considerable cost, lack of expertise and time involved. Consequently, the availability of genuine CPS dataset is limited. In response to this challenge, this paper leverages DoppelGANger, a Generative Adversarial Network (GAN) based model to generate synthetic CPS data using real four different CPS datasets across various domains: Water Distribution Testbed (WDT), Hardware in the Loop Industrial Control System (HAI), Gas pipeline, and Power system datasets. Furthermore, the synthetic datasets are compared to the real-world data through in-depth statistical analyses, visualization methods, and anomaly detection approaches. The difference in Silhouette scores across all CPS datasets is under 0.53, with the smallest difference of 0.071 in the HAI dataset and the largest difference of 0.525 in the Power System dataset. The Power system dataset exhibited the smallest difference in validation Mean Absolute Error (MAE), with values of 0.0184 for the real data and 0.0196 for the synthetic data. However, the correlation analysis revealed differences in feature relationships between real and synthetic datasets. Furthermore, the t-SNE visualizations for WDT demonstrated improved structural alignment between synthetic and real data as training epochs increased. Nevertheless, the alignment remained inconsistent for the HAI, Gas pipeline, and Power system datasets with increasing number of epochs. These observations highlight the need for more advanced synthetic data generation models within the realm of CPS to improve the fidelity and reliability of synthetic datasets.

Article Highlights

1. Underrepresentation of rare events in CPS synthetic datasets with larger batch size training.
2. T-SNE visualizations show structural misalignments with increasing complexity and size of CPS dataset.
3. DoppelGANger CPS synthetic datasets exhibit statistical deviations necessitating further refinements.

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1 Introduction

Academia and industry have experienced the influx of research on cyber-physical systems (CPS) in the last two decades making the subject area a force to reckon with [1]. CPS emerge from the integration of computation, networking, and physical processes, resulting in a transformative synergy amalgamating data from multiple sources, including sensors, databases, and external systems [2, 3]. Fig. 1 illustrates the architecture of CPS through two complementary views. Fig. 1a depicts the logical relationship between physical and cyber components in CPS. These components are arranged in a circular layout to emphasize their continuous interaction within the CPS environment. Fig. 1b presents the architecture of CPS across three primary layers: the Application Layer, the Networking Layer, and the Physical Layer. The physical layer contains sensors and actuators that interact directly with the physical environment, capturing data and performing actions based on received commands. This layer sends real-time sensor data to the networking layer, which includes devices like routers, switches, and servers that facilitate data and command flow within the CPS. Furthermore, security and configuration updates are managed at the physical layer to maintain the system's integrity and ensure optimal performance. The networking layer communicates upward to the application layer, where data is processed and stored, possibly involving cloud services. This layer analyzes the data and generates control commands which are sent back down to influence operations in the physical layer. Additionally, it manages configuration and security updates to adapt to evolving operational needs or threats.

Amongst the physical components of CPS are sensors, actuators making up of the control system. Sensors are devices that collect data from the physical environment. They measure parameters such as temperature, pressure, humidity, motion, and more. The digital component comprises of computers, processors, software, communication networks and data storage. Computers and processors process data from sensors, perform computations, and control actuators. Software can range from embedded control systems to high-level applications for analysis and visualization. Communication networks enable data exchange and coordination among components. Data storage systems, including databases and

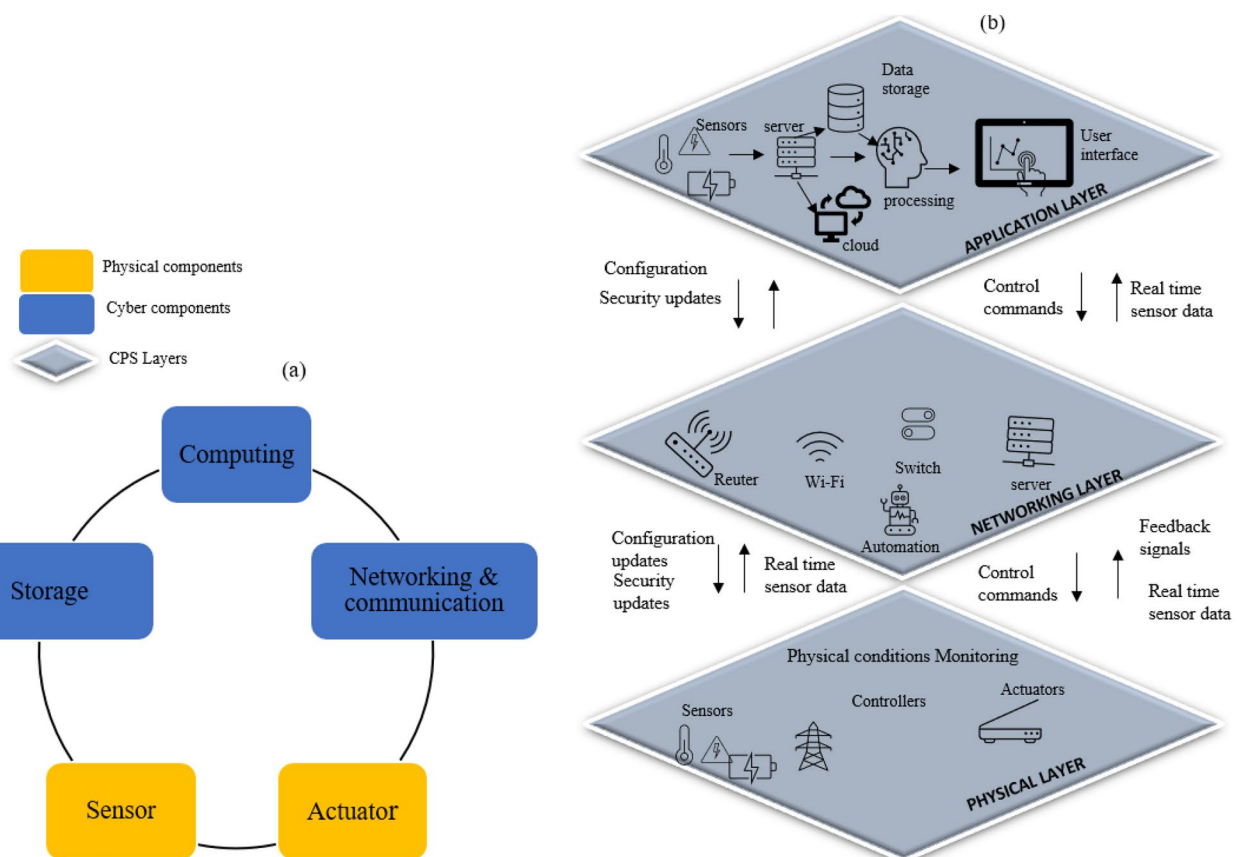


Fig. 1 Components of Cyber-Physical Systems and Layered Interactions

cloud storage, are used to manage and analyze this data. Human interaction plays a vital role in CPS. User interfaces and dashboards allow humans to interact with and provide input to the CPS. Human–Machine Interfaces are graphical interfaces that enable operators and users to interact with CPS by providing real-time data visualization and control capabilities.

The components of CPS draw from both the physical and digital realms while fostering human interaction, positioning CPS as a transformative field at the intersection of computation, networking, and the physical environment to enable predictive maintenance, anomaly detection, and autonomous control. CPS rely on accurate data collection, secure communication, and efficient decision-making algorithms to ensure reliability and safety. However, the interconnected nature of CPS also means vulnerability to cyber threats [4, 5] sensor failures [6], and operational inefficiencies [7], making robust data-driven solutions essential for improving resilience and performance [8, 9]. The development and deployment of intelligent CPS applications utilizing ML models play a crucial role in analyzing system behavior, predicting failures, and enhancing security [9].

Training ML models for CPS necessitate substantial amounts of data that accurately represent system operations and potential threats. However, the availability of public datasets in the domain of CPS is significantly limited due to various challenges, including time constraints, high costs, privacy concerns and the need for specialized technical expertise in data collection [10]. The time required to gather and prepare high-quality CPS data for training ML models is substantial [11]. This process entails the initial collection, extensive cleaning, preprocessing, and organizing of the data into a format suitable for use. The complexity and diversity of CPS environments amplify the time required for these tasks. Furthermore, the costs involved in collecting CPS data are often considerable. These environments typically feature advanced hardware and software, necessitating specialized equipment and tools for data monitoring and extraction. The continuous maintenance of these systems, as well as the infrastructure required for data storage and processing, add to the financial implications. Equally important is the technical expertise required for CPS data collection and processing. A profound understanding of both the cyber and physical elements of these systems is imperative. Professionals adept in both areas are crucial for not only effective data collection but also for ensuring the quality and relevance of the data for ML applications. This expertise is vital to navigate the complexities of CPS environments and to extract meaningful, usable data.

Given the challenges associated with collecting real-world CPS data, several synthetic data generation techniques have been explored. These include physics-based simulation approaches [12–15] and CPS rule-based modeling [16]. Among these, generative models offer a promising alternative due to their ability to learn complex, high-dimensional data distributions. Generative adversarial networks (GANs) have shown significant potential in producing high-quality synthetic datasets that preserve statistical and temporal properties of CPS environments. This study aims to fill the gap of CPS dataset scarcity by exploring the creation of synthetic cyber-physical datasets through DoppelGANger, an advanced GAN [17]. DoppelGANger was selected for CPS synthetic data generation due to its ability to capture complex temporal dependencies and preserve statistical properties in multivariate time-series data. To assess the performance of the synthetic dataset, we conducted comparative evaluations using four different CPS datasets across various domains: Water Distribution Testbed (WDT) [18], Hardware in the Loop Industrial Control System (HAI) [19], gas pipeline [20], and power system [21] datasets. We maintained consistent experimental settings across these domains to accurately evaluate the fidelity of the generated dataset. The contributions of this study are as follows:

1. Introduction of a synthetic CPS data generation and evaluation workflow, aiming to replicate the complexities of real-world CPS data across multiple domains.
2. Evaluation of the fidelity of synthetic datasets through visualization methods, statistical examination, and Silhouette score analyses.
3. Testing the synthetic datasets' effectiveness in simulating real-world CPS system anomalies and challenges leveraging anomaly detection approaches.
4. Comparative computational efficiency analysis between DoppelGANger and TimeGAN across different CPS domains.

The subsequent sections of the paper are organized as follows: section 2 presents literature review on generative adversarial networks developed to create synthetic image and tabular dataset. Section 3 provides a detailed description of all CPS datasets used, the preprocessing techniques employed, the model training process, and the techniques for evaluating the synthetic dataset generated and the real CPS dataset. Section 4 outlines the results of the evaluation techniques. The discussion of the results, limitations and practical applications of synthetic data in CPS are presented in section 5. The future research directions for synthetic dataset generation in CPS and conclusion are presented in Sections 6 and 7 respectively.

2 Related work

The application of GANs has proliferated across various domains since its introduction by Ian Goodfellow in 2014 [22]. Initially utilized for generating synthetic image datasets [22–28] the research scope of GANs expanded to include tabular datasets [29–39] and subsequently evolved to handle timeseries tabular datasets [40–48] evolving from traditional GANs to more advanced variants like Conditional GANs and Wasserstein GANs [35]. These techniques are applied not only in generating synthetic data but also in anomaly detection, data augmentation, and imputation tasks. The continuous evolution of GAN-based methods emphasize their versatility and potential for various applications in the field of AI and ML.

Several studies have explored domain-specific GAN architectures tailored for synthetic dataset generation within specialized application areas. Yu et al. [37] introduced a framework named SeqGAN to address the challenge of synthetic dataset generation. SeqGAN treats the data generator as a stochastic policy in Reinforcement Learning (RL), circumventing the issue of generator differentiation by employing direct gradient policy updates. The RL reward signal is derived from the evaluation of a GAN discriminator on a full sequence and is propagated back to the intermediate state-action steps through Monte Carlo search. A drawback of this approach lies in its complexity. Implementing and training a SeqGAN model demands knowledge and understanding of both reinforcement learning and generative adversarial networks. This intricacy poses a challenge for individuals lacking expertise in these domains. Arkangil et al. [38] developed a framework to generate a synthetic population with trip chains using a combination of GAN with recurrent neural network (RNN). GAN training already faces challenges like mode collapse and convergence difficulties and integrating an RNN into the GAN framework can amplify these issues. Lederrey et al. [39] introduced the Directed Acyclic Tabular GAN (DATGAN) to address these limitations by modeling the generator's structure for synthetic tabular data generation. This approach allows the interactions between variables to be specified explicitly using a Directed Acyclic Graph (DAG). However, the influence of multiple variables in a cluster were not considered in modeling. In [35], the authors proposed a Wasserstein GAN-based approach to synthesize tabular datasets focusing on adding additional loss function for improved performance. In the financial domain, Dannels et al. [44] employed a GAN to generate synthetic time series data for treasury yields and recession indicators, leveraging a dataset from the Federal Reserve Economic Data (FRED) to support short-term forecasting. A limitation of the study is that it relied on a single dataset, potentially restricting the model's ability to generalize across different economic conditions. Moreover, Asre and Anwar. implemented TimeGAN energy consumption dataset generation for five States of Australia [49]. Recently, Alshantti et. al proposed CastGAN, a Cascaded Generative Adversarial Network for synthetic tabular dataset generation [50]. The authors of [51] and [52] provided reviews of GAN design research focused on tabular data generation.

Beyond GANs, researchers have explored various advanced neural architectures for generating synthetic sequential data for specific domains. Zhang et al. [40] applied Long-Short-Term-Memory (LSTM) to generate synthetic well logs using existing log data, addressing missing logging information without increasing costs using LSTM. The study was specific to certain types of well log data or contexts, potentially limiting its applicability to broader datasets. Alzantot et al. [36] proposed a method that utilized a generator model comprising multiple LSTM networks and a Mixture Density Network (MDN) to synthesize sensory data. The authors also introduced an LSTM-based discriminator model to distinguish between real and synthetic data. Zhang and colleagues [41] presented the Conditional Probabilistic Auto-Regressive (CPAR) model, a neural network-based approach, within a Synthetic Data Vault (SDV) framework, designed specifically for generating sequential data. While the study introduces a new metric Multi-Sequence Aggregate Similarity (MSAS) to compare synthetic and real data, the limitations of this metric were not discussed. However, the study did not assess the metric's robustness or compare CPAR to other sequential GANs. This limited scope of comparison might not provide a comprehensive understanding of the Sequential SDV model's performance in comparison to existing sequential generative models. Padhi et al. [42] introduced Hierarchical Tabular BERT and Tabular GPT for modeling multivariate times series leveraging Transformers. Meiser et al. [43] introduced SynTiSeD, the Synthetic Time Series Data Generator, a simulation tool driven by multiple agents that produces realistic energy data derived from real-world sources. SynTiSeD not only generates typical energy datasets but also facilitates the creation of crucial scenarios vital for service development, scenarios that are challenging to replicate in real-world situations.

Research on creating synthetic datasets for CPS is limited, due to the complex nature of these systems. Hasan et al. [45] proposed a novel Intrusion Detection System based on GAN to tackle issues related to imbalanced and missing data by generating synthetic samples through GAN technology. The efficacy of this model was evaluated using the

NSL KDD-99 dataset. The NSL-KDD dataset consists solely of network traffic data, including TCP/IP headers, various attack scenarios, and regular network activities. It does not incorporate any physical data, making it unsuitable for representing the complexities inherent in CPS. As far as the authors are aware, few studies have explicitly focused on generating synthetic CPS datasets beyond network traffic data. Zimmering et al. [46] introduced an approach, defining a unique data generation model that replicates the complex patterns inherent in CPS. Rooted in well-established system theory, this model offers the flexibility to generate datasets with diverse complexities. These datasets served as a pivotal tool to enable a systematic and unbiased comparison of selected ML methods. The study assessed the performance of the proposed data generation process employing techniques such as Self-Organizing Map (SOM), One-Class Support Vector Machine (OCSVM), and LSTM in anomaly detection. The study's limitations include a lack of real-world validation, simplistic model assumptions that might not capture the complexities of actual CPS environments. In [53], the authors applied the TimeGAN framework to generate synthetic cyber-physical water data. They conducted experiments using two real-world datasets: the Secure Water Treatment (SWaT) dataset from the University of Singapore and the BATADAL competition dataset.

To address the current deficiencies in the literature, this study employs DoppelGANger, a GAN-based model specifically designed for structured time-series data synthesis as outlined by Lin et al. [17] to create four CPS datasets across different domains. These datasets are derived from genuine data collected from various real-world CPS environments including Water Distribution Testbed [18], HAI-22-04 [19], Gas Pipeline [20], and Power System [21] datasets. The generated datasets are validated through statistical analysis, visualization techniques, and anomaly detection to ensure their applicability to real-world CPS research. The primary goal of this approach is to foster a deeper understanding of CPS behaviors and patterns.

3 Methodology

The methodology of this study commences with the acquisition of four CPS datasets from diverse sources. These datasets are essential for understanding the complex interactions within CPS across various sectors, including WDT [18], HAI-22-04 [19], Gas Pipeline [20], and Power System [21] datasets. Each dataset offers unique insights into the specific attributes and operational behaviors of CPS. These CPS datasets undergo data preprocessing steps such as cleaning, file merging, label encoding, and normalization. The preprocessed data is then fed into the DoppelGANger model for synthetic data generation model training to replicate real data characteristics. The generated CPS data is evaluated and validated to ensure fidelity, followed by fine-tuning to optimize model performance. Following model training, the generated synthetic data is evaluated and validated against real data using statistical and machine learning-based comparison metrics to assess fidelity and utility. The Synthetic CPS data generation and performance evaluation workflow is shown in Fig. 2.

3.1 Datasets

Four different CPS datasets across various domains namely: WDT [18], HAI-22-04 [19], gas pipeline [20], and power system [21] datasets were utilized. Detailed description based on number of features and samples are presented in Table 1.

Fig. 2 Synthetic CPS Data Generation and Evaluation Workflow

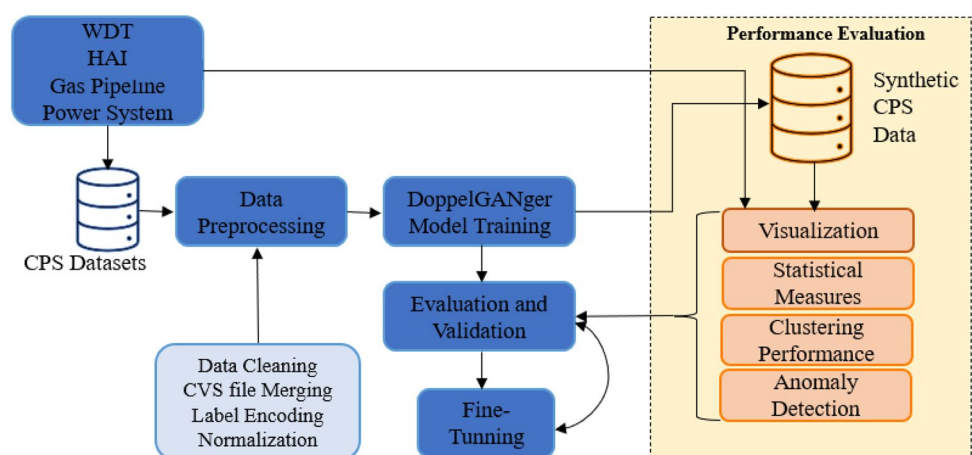


Table 1 Summary of CPS datasets

Dataset	Number of features	Number of samples
WDT [18]	52	10,902
HAI-22-04 [19]	88	1,365,602
Gas Pipeline [20]	21	20,778
Power System [21]	128	72,072

3.1.1 WDT dataset

Faramodi et al. created the WDT CPS dataset, which includes eight tanks equipped with solenoid valves, pumps, pressure sensors, and flow sensors [18]. The dataset encompasses simulation data spanning a 2-hour period under normal operating conditions, as well as various anomalies caused by both cyber and physical attacks. Physical attacks are attacks against the physical elements such as sensors and actuators whilst cyber attacks are attacks against hosts or communication links as described by the authors. This comprehensive dataset has been made publicly accessible via IEEE Dataport to support and facilitate research in the field. According to the authors, attack_1, attack_2 and attack_3 refer malicious network traffic. Table 2 presents the WDT CPS dataset description while Table 3 provides the physical and network feature descriptions along with their corresponding data types.

Table 2 Description of WDT Dataset

Category	Network	Network resampled	Physical
Attack 1	5527409	2420	2420
Attack 2	5159469	2096	2104
Attack 3	5862547	1252	1254
Attack 4	5522490	1711	1717
Normal	7757289	3425	3428
Total	29829204	10904	10923

Table 3 Description of Features of WDT Dataset

Feature	Data type	Description
<i>Physical</i>		
Timestamp	Datetime	Date of acquisition
Tank_1 to Tank_8	Numeric	Pressure sensor value of tank 1 to tank 8
Pump_1 to Pump_6	Boolean	State of pump 1 to pump 6
Flow-sensor-1 to Flow_sensor_4	Numeric	Flow sensor values
Valv_1 to Valv_22	Boolean	State of the 22 solenoid valves
<i>Network</i>		
Timestamp	Datetime	Date of acquisition
Src IP address	Categorical	Source IP address
Dst IP address	Categorical	Destination IP address
Src MAC address	Categorical	Source MAC address
Dst MAC address	Categorical	Destination MAC address
Src Port	Numeric	Source port
Dst Port	Numeric	Destination port
Proto	Numeric	Protocol
TCP flags	Numeric	Flags
Payload size	Numeric	Size of payload
MODBUS code	Categorical	MODBUS function code
num_pkts_src	Numeric	No. of packets of the same source address in the last 2 seconds
num_pkts_dst	Numeric	No. of packets of the same destination address in the last 2 seconds

Table 4 Description of Features of HAI-22.04 Dataset

Feature range	Data type	Description
Timestamp	Datetime	The time at which the data was recorded
P1_B2004 to P1_TIT03	Numeric	Various sensor readings and indicators for section P1
P2_24Vdc to P2_VTR04	Numeric	Control signals and measurements for section P2
P3_FIT01 to P3_PIT01	Numeric	Flow and pressure indicators for section P3
P4_HT_FD to P4_ST_TT01	Numeric	High-temperature and other critical parameters for section P4

Table 5 Description of Features of Gas Pipeline Dataset

Feature	Data type	Description
Address	Categorical	Unique address identifier for each pipeline segment
ControlScheme	Categorical	Type of control scheme implemented
CRC	Numeric	Cyclic redundancy check value for error detection
DataLength	Numeric	Length of the data transmitted
FunctionCode	Categorical	MODBUS function code used in communication
InvalidDataLength	Boolean	Flag for data length mismatches
InvalidFunctionCode	Boolean	Flag for unrecognized function codes
PIDCycleTime	Numeric	Cycle time of PID controller
PIDDeadband	Numeric	Deadband value in PID controller settings
PIDGain	Numeric	Gain parameter of the PID controller
PIDReset	Numeric	Reset rate of the PID controller
PipelinePSI	Numeric	Pressure inside the pipeline in PSI
SetPoint	Numeric	Target set point for control measures
TimeInterval	Numeric	Time interval for data capture
DeltaPIDCycleTime	Numeric	Change in PID cycle time from previous measurement
DeltaPIDDeadband	Numeric	Change in PID deadband from previous measurement
DeltaPIDGain	Numeric	Change in PID gain from previous measurement
DeltaPIDRate	Numeric	Rate of change in PID parameters
DeltaPIDReset	Numeric	Change in PID reset rate from previous measurement
DeltaPipelinePSI	Numeric	Change in pipeline pressure from previous measurement
DeltaSetPoint	Numeric	Change in control set point from previous measurement

3.1.2 HAI-22.04 Dataset

Multiple versions of the HAI dataset exist. In this instance, we are focusing on HAI-22.04, released in 2022 [19]. HAI-22.04 comprises of detailed records from industrial automation systems, with 88 columns and 1,365,602 samples, highlighting interactions in high automation environments. HAI-22.04 includes time-series data from flow meters, pressure indicators, temperature sensors, and actuator control signals, enabling anomaly detection and system monitoring. The dataset is structured into four main sections (P1 to P4), each representing a distinct subsystem of the industrial process. Table 4 outlines the features and corresponding data types of the HAI dataset.

3.1.3 Gas pipeline dataset

The gas pipeline dataset was developed in two phases using a lab-scale system. Initially, a simulation of 35 different types of attacks and HMI user interactions was conducted via AutoIT within the designated framework. Following this, all events were documented. This dataset primarily consists of traffic data captured between the Master Terminal Unit (MTU) and the slave Remote Terminal Unit (RTU), encompassing both normal and malicious events. The dataset contains data related to the operations of gas transmission systems, including 21 features and 20,778 samples, which helps in analyzing pipeline integrity and operational safety [20]. A detailed feature description is provided in Table 5.

3.1.4 Power system dataset

We utilized a publicly accessible power system dataset collaboratively developed by Mississippi State University and Oak Ridge National Laboratory [21]. This miniaturized version of the power system includes two generators (G1, G2), four Intelligent Electronic Devices (IEDs) labeled R1-R4, and four circuit breakers (BR1-BR4). The IEDs control the activation and deactivation of these breakers, and two circuits are formed using breaker pairs (BR1-BR2 and BR3-BR4). Additionally, the field system connects to a control room via a substation switch and a Power Distribution Center (PDC). The dataset is available in 15 segments and comes in three formats: binary, ternary, and multi-class [21]. For our research, we employed the binary version, which includes 128 features derived from four Phasor Measurement Units (PMUs) event and control logs and 72,072 samples used to study energy distribution and grid stability. Table 6 summarizes the Power System Dataset features.

3.2 Data preprocessing

The WDT real-world cyber-physical dataset underwent preprocessing steps, which included checking for missing, duplicated, and NaN values. The implementation was carried out using Python programming language. Additionally, the network dataset was resampled, creating samples at one-second intervals during the experiment, achieved through a Python function. This resampling facilitated the integration of network and physical data, allowing for the analysis of a comprehensive CPS. Categorical features were transformed using label encoding techniques owing to its effectiveness with extensive datasets and also to enable models to process information swiftly. Furthermore, timestamp column was converted to datetime format to capture temporal dependencies during training. Lastly, the dataset was normalized. The resulting dataframe comprised of Ten thousand, nine hundred and two (10902) samples, encompassing fifty-two (52) network and physical features.

The HAI–22.04 dataset, consisting of six training CSV files and four testing CSV file also underwent a thorough preprocessing process. The files were combined into a single dataset. This combined dataset was then cleaned to remove duplicates and missing values. After the cleaning process, the dataset was normalized, resulting in a total of one million three hundred and sixty-five thousand, six hundred and two (1,365,602) rows and 88 columns. The training CSV files from the HAI–22.04 CPS dataset exclusively contain samples representing the system’s normal operational state. As a result, when these files were combined, the ratio of attack samples to normal samples was extremely low. This imbalance adversely affected the data generation model’s ability to accurately replicate attack scenarios. To address this issue, the second testing CSV file was chosen for generating the synthetic dataset because it included a better distribution of data with 125,177 normal samples and 4,423 attack samples.

The preprocessing of the gas pipeline dataset involved several steps to ready the data for model training. All individual CSV files were merged into a single dataframe. Specific columns deemed irrelevant for further analysis, such as ‘CommandResponse’, ‘ControlMode’, ‘PIDRate’, ‘PumpState’, and ‘SolenoidState’, were removed. Duplicate rows were also eliminated from the combined dataframe to enhance data quality. Furthermore, several columns were transformed into boolean numeric values (0 or 1) to simplify the representation of binary categorical data. Non-numeric columns were converted to numeric formats using the LabelEncoder, and the entire dataset underwent normalization to ensure uniformity in data scale and range. The final dataframe contained Twenty thousand, seven hundred and seventy-eight (20,778) rows, twenty-one (21) features and a target column.

Table 6 Description of Features of the Power System Dataset

Feature range	Data type	Description
R1-PA1:VH to R1:S	Numeric	Voltage and status measurements across phase 1
R2-PA1:VH to R2:S	Numeric	Voltage and status measurements across phase 2
R3-PA1:VH to R3:S	Numeric	Voltage and status measurements across phase 3
R4-PA1:VH to R4:S	Numeric	Voltage and status measurements across phase 4
Control panel log1 to log4	Integer	Log data from control panel operations
Relay log1 to log4	Numeric	Log data from relay operations
Snort log1 to log4	Numeric	Log data from network security monitoring

For the power system CPS dataset, fifteen (15) csv files were combined into a dataframe and the preprocessing included cleaning, handling of missing values and normalization. The final dataframe had seventy-two thousand and seventy-two (72,072 rows), 128 features and a target column.

3.3 DoppelGANger

GANs are a class of AI algorithms used in unsupervised ML, implemented by a system of two neural networks contesting with each other in a zero-sum game framework. This technique was introduced by Ian Goodfellow and his colleagues in 2014 [22]. GANs are composed of two models namely the generator and the discriminator, which are trained simultaneously by an adversarial process. Fig. 3 presents the basic architecture of a GAN model.

The generator takes random noise as input and generates data that is similar to the training data while the Discriminator evaluates the data for authenticity, thus distinguishes between real (training) data and fake (generated) data from the generator. GANs have been used for various tasks such as image generation, photo realistic images, super-resolution, style transfer, and more. The objective function of a GAN can be described using the following equation:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

G is the generator mode minimizes the objective function above. D represents the discriminator model. x is a real sample from the data distribution $p_{data}(x)$. z is a random noise vector sampled from a prior noise distribution $p_z(z)$. $D(x)$ is the discriminator's estimate of the probability that the real data instance x is real. $G(z)$ is the output of the generator given noise z . $D(G(z))$ is the discriminator's estimate of the probability that a fake instance produced by the generator G is real.

DoppelGANger is a GAN-based model specifically tailored to create complex synthetic multivariate time series data. This model can produce realistic and diverse synthetic sequences that mimic the characteristics of a given dataset [17]. The architecture leverages LSTM generators (a category of RNNs) to encapsulate and model temporal dependencies within the data. The innovative aspect of DoppelGANger is its joint employment generators to produce synthetic attributes, also known as metadata, that accompany the synthetic time series. In its operational flow, an Multilayer Perceptron (MLP) initially synthesizes metadata which serve as input for another MLP as depicted in the architecture in Fig. 4. This subsequent MLP is responsible for the generation of metadata bounds that is the minimum and maximum values that are utilized during the normalization process. These metadata are then fed into the LSTM generator to generate the synthetic time series.

Moreover, DoppelGANger integrates an auxiliary discriminator within its architecture. The purpose of this component is to effectively manage the disparate dimensionalities and data types that are being synthesized, to ensure the generated time series and their corresponding metadata are coherent and mimic the statistical properties of real-world data.

3.4 Model training

The model was trained leveraging the Gretel DGAN model API which provides DoppelGANger the generative data Model for time-series CPS data. This versatile model accommodates dynamic features over time, stable attributes, categorical variables, and is compatible with a wide array of time sequence. Source and destination IP addresses together with the

Fig. 3 The Achitecture of a GAN

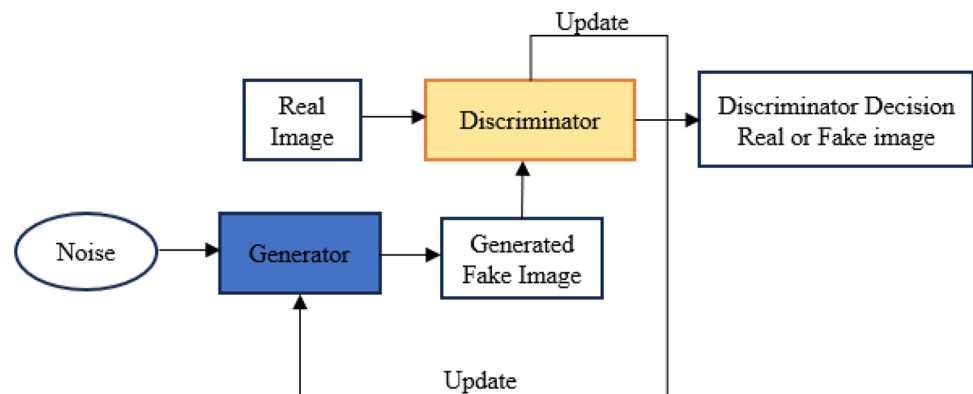
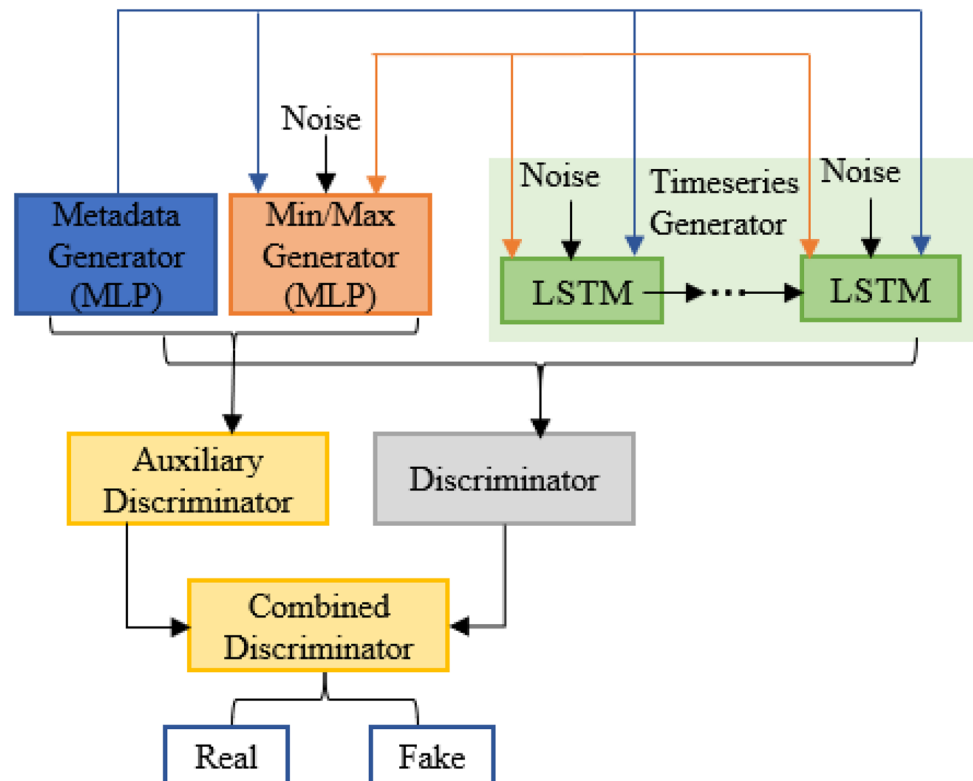


Fig. 4 The Architecture of DoppelGANger

source and destination MAC addresses were treated as discrete categories within the data parameters. Additionally, the timestamp column was designated as the time parameter in the dataset.

The GAN generator network was configured with four (4) LSTM layers, while the GAN discriminator network employed six (6) layers as presented in Table 7. All other model parameters remained at their default values. Fine-tuning in this context refers to the systematic adjustment of hyperparameters to optimize the model's ability to capture the underlying statistical properties of CPS data. After fine-tuning and retraining based on evaluation results, the DGAN model underwent training for 100, 250, 500, and 1000 epochs. This extensive training allowed for a detailed comparison between the generated synthetic data and the authentic CPS dataset, aiming to assess their proximity. The iterative adjustments made during training underscore the dedication to refining the generative data model, enhancing its accuracy and reliability in generating synthetic time-series data. Throughout the training process, the model's hyperparameters were fine-tuned to acquire optimal settings, resulting in superior performance. The experiments were performed using the following hardware and software platforms: Intel(R) Core (TM) i7-8550U CPU, 4 Core(s), 8 Logical Processors, Windows 10 Professional (64-bit), NVIDIA GeForce MX 150, NVIDIA CUDA 10.2, Python 3.8, and Google Colab with NVIDIA A100 Tensor Core GPU.

3.5 Evaluation techniques for real and synthetic CPS datasets

In order to ensure the CPS synthetic datasets accurately represents the characteristics of the real data, the proximity between was evaluated using visualization techniques, statistical measures, correlation analysis, clustering analysis and anomaly detection approaches.

Table 7 DoppelGANger Model Configuration Parameters

Parameter	Configuration
Batch_size	1000
Attribute_num_layers	6
Feature_num_layers	4
Parameters are set before model training	

3.5.1 Visualization techniques

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a ML algorithm used for dimensionality reduction and visualization of high-dimensional data in lower-dimensional space, typically 2D or 3D. This technique is useful for visualizing the complex CPS datasets with many features because t-SNE captures the underlying patterns and structures of the data, making it easier to interpret [54]. t-SNE was selected because, unlike many other dimensionality reduction techniques, it is non-linear and captures complex non-linear relationships in the WDT, HAI, Gas pipeline and Power system CPS datasets.

3.5.2 Statistical measures

The mean is a measure of central tendency in a dataset while the standard deviation is a measure of the dispersion or spread of a dataset, quantifying how much the values in a dataset deviate from the mean. A low standard deviation indicated that the values tend to be close to the mean, while a high standard deviation suggested that the values are spread out over a wider range. Comparing the means of corresponding variables in synthetic and real data provided insights into whether they exhibited similar central tendencies. Means closer in values suggested synthetic data was similar to the real data. Comparing the standard deviations helped assess the spread or variability of the data points around the mean. Similar standard deviations indicated comparable variability in the synthetic and real datasets. However, the mean and standard deviation alone provide a limited perspective. To statistically assess the similarity between real and synthetic CPS datasets, the t-test and Levene's test was used [55, 56]. The t-test compares the means of each feature across the CPS datasets and evaluates whether any observed differences are statistically significant. A high t-statistic coupled with a low p-value (typically below 0.05) indicates that the difference in means between the real and synthetic data is statistically significant [55]. The Levene's test is used to assess the equality of variances between two datasets. While the t-test focuses on the differences in means, Levene's test evaluates whether the variability or spread of the data differs between real and synthetic datasets. If the variances are significantly different, the synthetic data does not replicate the real data's variability [56].

3.5.3 Correlation analysis

Karl Pearson introduced the Pearson correlation coefficient in 1895, providing a mathematical method for quantifying the strength and direction of linear relationships between two variables [57]. Correlation analysis is a statistical technique used to evaluate the strength and direction of the linear relationship between two variables. It helps to identify how closely related two variables are, with the correlation coefficient (often denoted as r) indicating the strength and direction of this relationship. The coefficient ranges from -1 to 1 , where a value of 1 indicates a perfect positive correlation, meaning that as one variable increases, the other also increases proportionally; a value of -1 indicates a perfect negative correlation, where as one variable increases, the other decreases proportionally; and a value of 0 indicates no linear relationship between the variables. A correlation above 0.9 is generally considered very high and can interfere with ML models, especially logistic regression [58, 59]. High correlation, or multicollinearity, can make it difficult for the model to determine the individual contribution of each feature, leading to unstable or inflated coefficients.

3.5.4 Clustering performance analysis

The Silhouette Score introduced by Rousseeuw assesses how well-separated and cohesive clusters are by measuring the similarity of a point to its assigned cluster compared to other clusters [60]. It ranges from -1 to 1 , where higher values indicate better clustering with a score close to 1 meaning that the sample is well-clustered, whereas values near 0 imply that the sample is on the boundary between clusters, and negative values indicate misclassification. The clustering performance analysis has been utilized in works like [61, 62], and [63]. The Silhouette score is defined by [60] as:

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (2)$$

where:

$S(i)$ is the silhouette score for a data point i .

$a(i)$ is the average intra-cluster distance (the mean distance between i and all other points in the same cluster).
 $b(i)$ is the average nearest-cluster distance (the mean distance between i and all points in the nearest neighboring cluster).

3.5.5 Unsupervised anomaly detection

The decision to use the isolation forest, originally introduced by Liu and colleagues [64] for unsupervised anomaly detection, is rooted in its proven effectiveness with high-dimensional datasets. Moreover, the scalability, interpretability, and flexibility of the isolation forest algorithm make it adept at capturing intricate outliers effectively. Within the isolation forest framework, features in the dataset undergo random selection and recursive partitioning. This process aimed to pinpoint anomalous data points that reside farther away from the typical observations within the feature space. In this study, dimensionality reduction of the feature matrix is performed leveraging principal component analysis (PCA) [65] before training an isolation forest model based on the reduced features. The dataset is divided into training and testing subsets to facilitate model evaluation. A hyperparameter grid is defined for the Isolation Forest. Performance is assessed on the test set by predicting anomalies and calculating metrics including weighted F1 score, precision, and recall. The detailed algorithm is presented in Algorithm 1.

Algorithm 1 PCA-Isolation Forest with Hyperparameter Tuning

Inputs: Feature matrix X , Target variable y
Output: Performance metrics: Precision, Recall, F1-Score, Accuracy

- 1: Apply PCA to reduce dimensionality of X
- 2: Split reduced data into training and testing sets
- 3: Define hyperparameter grid: $n_{estimators}$, $max_{samples}$, $contamination$, $max_{features}$
- 4: Perform grid search with cross-validation to find optimal parameters
- 5: Train Isolation Forest using optimal parameters on training set
- 6: Predict on test set using trained model
- 7: Map prediction labels (if necessary)
- 8: Compute evaluation metrics: precision, recall, F1-score, accuracy
- 9: Return best hyperparameters

3.5.6 Anomaly detection based on deep learning - quantile stacked LSTM.

Deep learning anomaly detection refers to the use of DL techniques to identify unusual patterns in data that do not conform to expected behavior. This approach is effective in handling the complex and high-dimensional datasets of CPS. Anomaly detection based on LSTM networks is a DL technique effective for identifying anomalies in time-series data. LSTM networks, introduced by Hochreiter [66] are well-suited for processing, predicting, and classifying CPS time-series data given their ability to remember long-term dependencies. Quantile Stacked LSTM (QS-LSTM) network is adopted for this study of anomaly detection due to the temporal dependency nature of the model and the ability to filter out noise. Utilizing stacked LSTM layers improved the quality of representation learning from CPS sequential data. Quantile regression initially proposed by Koenker [67] is used to predict the 90th percentile of the MAE loss values from the training data to serve as the threshold for anomaly detection. The model architecture comprises multiple LSTM layers with decreasing units, capturing complex temporal dependencies, followed by a dense output layer. Training involves minimizing the mean absolute error (MAE) between predicted and actual values. The model forecasts on the training set to compute MAE losses after training, which are then used to train a quantile regression model, establishing thresholds for different quantiles. During evaluation, the model calculates MAE losses on test data; instances where this loss exceeds the predefined quantile threshold are flagged as anomalies, indicating significant deviations from the learned normal patterns. This approach effectively captures the underlying distribution of time-series data. The threshold obtained from this quantile served as a reference point for identifying anomalies in the test set. The detailed algorithm is presented in Algorithm 2. The formula for MAE is provided below.

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (3)$$

Y_i are the actual observed values, $\hat{Y}_i$ are the predicted values and n represents the total observations in the dataset.

Algorithm 2 Quantile Stacked LSTM for Anomaly Detection

Inputs: Feature matrix X , Target variable y

Output: Performance metrics: MAE loss

- 1: Data Preprocessing
 - 2: Prepare sequences for LSTM:
 - 3: Define n_{past} and n_{future}
 - 4: Create sliding window sequences of inputs X and targets y
 - 5: Split data into training and validation sets
 - 6: Define stacked LSTM model architecture
 - 7: Add LSTM layers
 - 8: Add Dense layer with linear activation
 - 9: Compile model with MAE loss and Adam optimizer
 - 10: Train model on training set
 - 11: Compute training MAE loss L_{train}
 - 12: Train quantile regression model on L_{train} to determine threshold T
 - 13: Predict validation MAE loss L_{val}
 - 14: Classify anomalies:
 - 15: If $L_{val} > T$: Anomaly
 - 16: Else: Normal
-

4 Results

4.1 Results comparison based on t-SNE visualization

The comparison of WDT synthetic data is conducted across different epochs, as illustrated in the visualizations including data generated at 100, 250, 500, and 1000 epochs. The t-SNE visualization of each epoch for the generated synthetic and real CPS data is shown in Fig. 5. In the Fig. 5a, representing visualizations from running 100 epochs, it is evident that the clusters in the synthetic dataset do not closely resemble those in the real dataset. The patterns of data points created in the synthetic data after 100 iterations are significantly different from the patterns observed in the real dataset suggesting the algorithm used to generate the synthetic data has not yet converged to accurately replicate the underlying structures of the real CPS data. In Fig. 5d, The synthetic dataset, after one thousand (1000) epochs, starts to show patterns and clusters that align closely with the patterns observed in the real dataset. The synthetic data generated after one thousand (1000) epochs demonstrates improved accuracy compared to earlier iterations.

An equivalent number of samples from the WDT dataset were drawn from the HAI-22.04, Gas Pipeline, and Power System datasets. Despite this uniformity in sampling, the visual outputs of the synthetic datasets did not closely resemble those of the corresponding real datasets. These discrepancies in the HAI-22.04, Gas Pipeline, and the Power System datasets after 1000 epochs are illustrated in Figs. 6, 7 and 8 respectively. Furthermore, Table 8 displays the duration in seconds used for training the DoppelGANger model across the different CPS datasets.

4.2 Result comparison based on statistical measures

Table 9 presents a statistical analysis of the numeric features in the real and synthetic WDT CPS data, including the results of t-tests for comparing means and Levene's tests for comparing variances. The p-values of Tank_1, Tank_3, Tank_4, Tank_5, Tank_6, Tank_7 are extremely low (much less than 0.05), indicating statistically significant differences

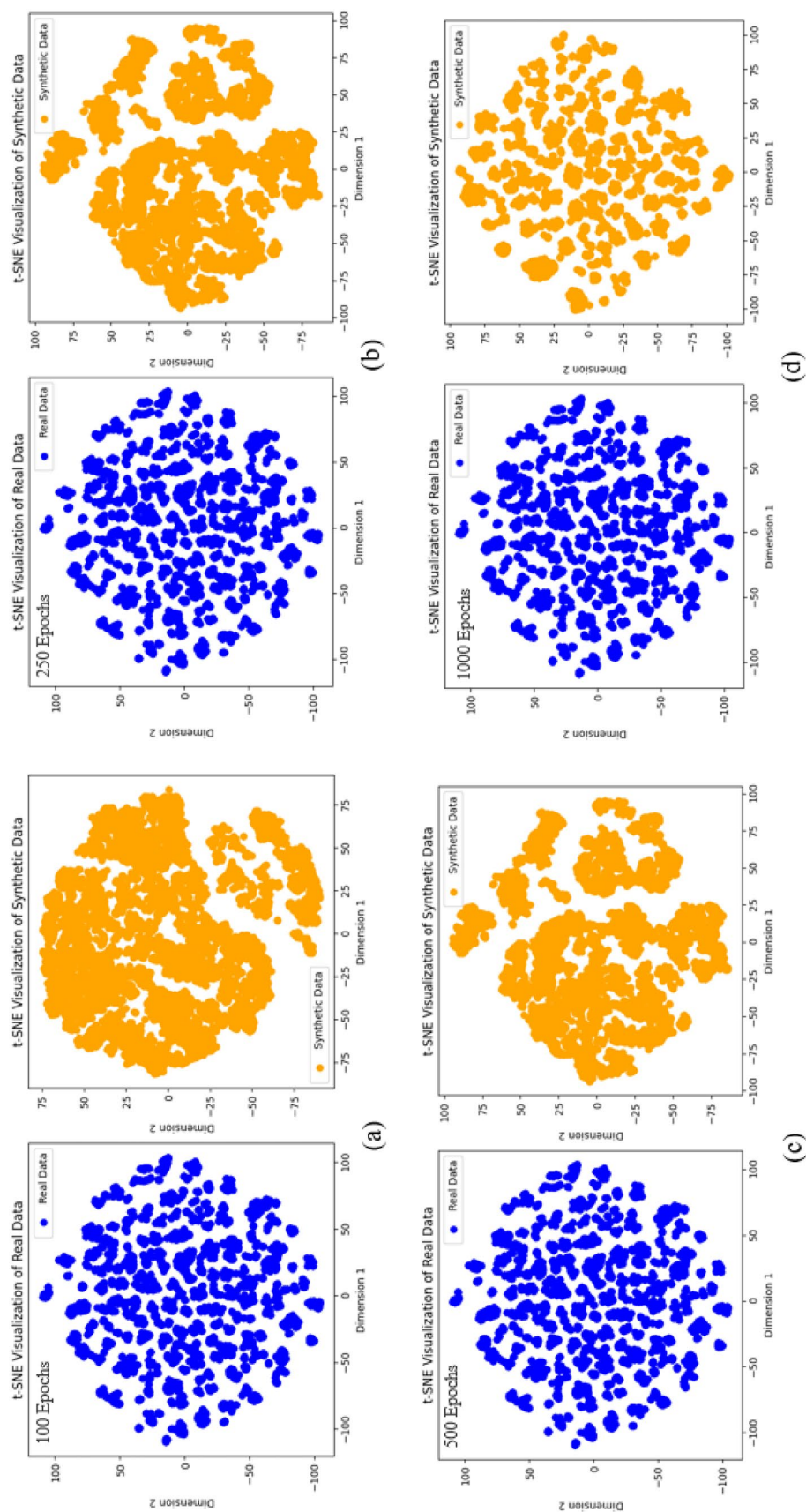


Fig. 5 t-SNE Visualizations of Real and Synthetic WDT at Different Training Stages. (a) Visualizations at 100 epochs, (b) at 250 epochs, (c) at 500 epochs, and (d) at 1000 epochs

Fig. 6 t-SNE Visualization of Real and Synthetic HAI CPS Data

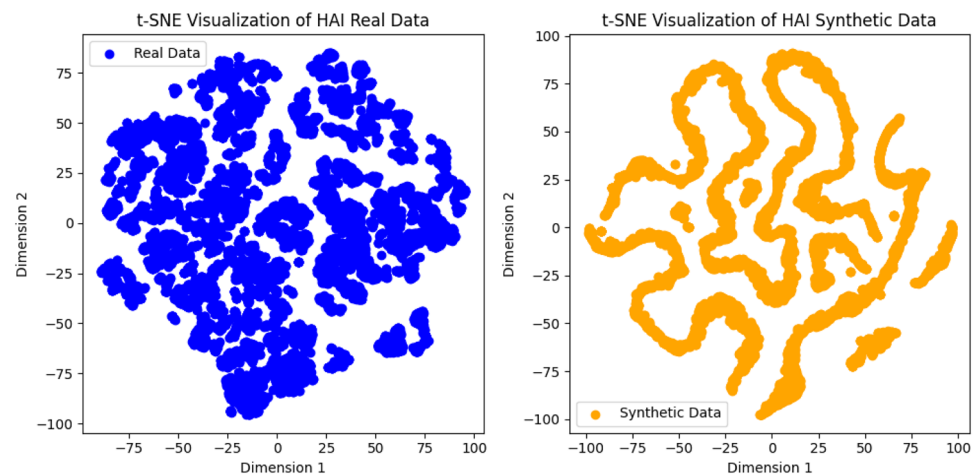


Fig. 7 t-SNE Visualization of Real and Synthetic Gas Pipeline CPS Data

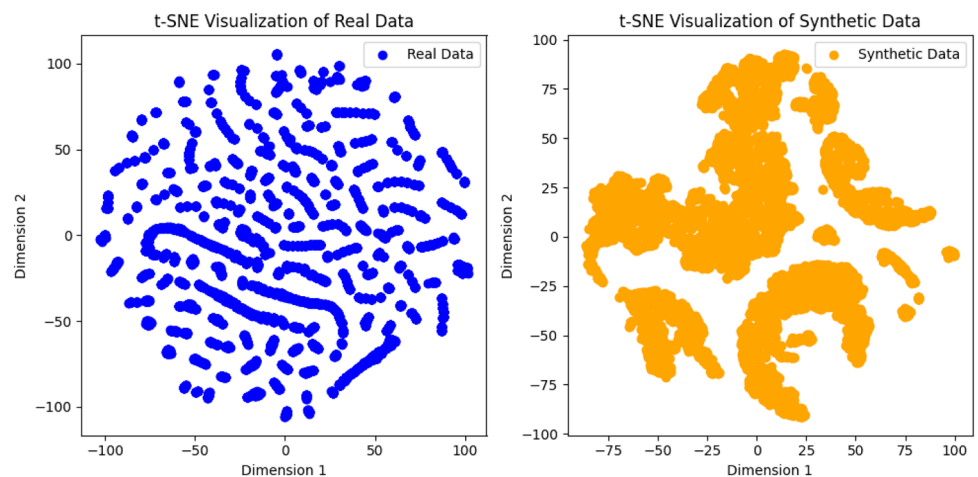
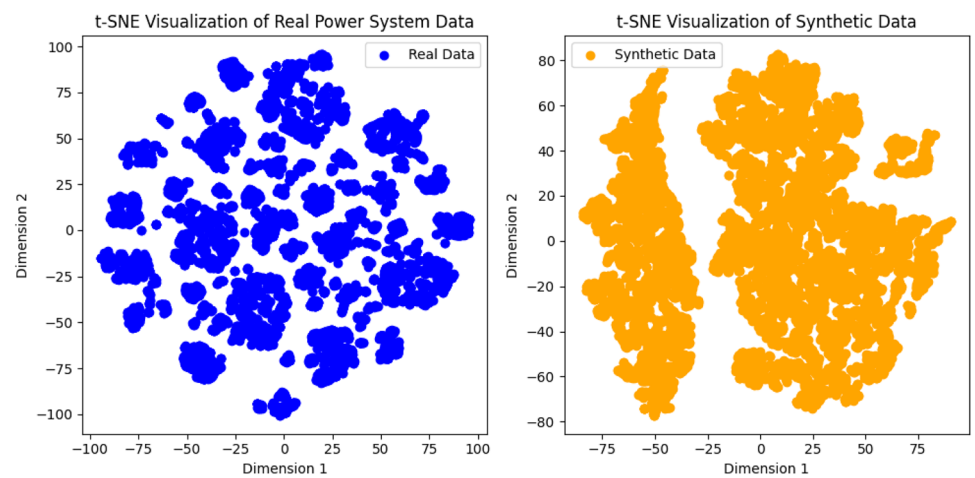


Fig. 8 t-SNE Visualization of Real and Synthetic Power System CPS Data



between the means of the real and synthetic WDT data. The synthetic data for these features does not perfectly replicate the real data, despite potentially capturing the overall trend. However, the p-values for Tank_2 and Tank_8 are greater than 0.05 (0.191 for Tank_2 and 0.809 for Tank_8) which indicates no statistically significant differences between the means of the real and synthetic data. The synthetic data for these features closely matches the real data, making the synthetic data a good representation of the real data for these specific features. The t-test results show

Table 8 Training Time for Different and Number of Epochs for Different CPS Datasets

Dataset	Number of epochs	Time (s)
WDT	1000	3446.75
HAI	1000	13796.21
Gas pipeline	1000	8090.44
Power system	1000	5257.16

Table 9 Statistical Analysis of Real and Synthetic WDT CPS Data

Features	Mean (Real)	Mean (Synthetic)	T-Statistic	P-value (T-Test)	Std Dev (Real)	Std Dev (Synthetic)	Levene statistic	P-value (Levene Test)
Tank_1	0.359	0.332	6.261	3.903e-10	0.334	0.302	0.406	5.249e-01
Tank_2	0.448	0.442	1.307	1.913e-01	0.342	0.336	2.621	1.071e-01
Tank_3	0.173	0.132	18.925	3.072e-79	0.180	0.137	3.107	7.951e-02
Tank_4	0.233	0.260	-6.751	1.507e-11	0.282	0.308	0.238	6.264e-01
Tank_5	0.439	0.513	-16.067	9.395e-58	0.334	0.346	2.025	1.564e-01
Tank_6	0.241	0.173	25.228	1.923e-138	0.218	0.178	11.225	9.663e-04
Tank_7	0.226	0.186	12.695	8.463e-37	0.249	0.215	2.903	8.997e-02
Tank_8	0.330	0.329	0.241	8.093e-01	0.358	0.243	13.842	2.590e-04

Water level in tanks is measured in meters

that Tank_2 and Tank_8 closely replicate the real data while Tank_1, Tank_3, Tank_5 show significant differences. This variation presents the need for refinement to ensure that all features of the real data are accurately captured. Levene's test results indicate that most features do not have significant differences in variances between the real and synthetic data except for Tank_6 and Tank_8. The synthetic data generally captures the variability of the real data well although there are some discrepancies in specific features of the WDT dataset.

The statistical comparison between the numeric features of the real and synthetic HAI dataset, as presented in Table 16 of Appendix A, reveals significant differences across most features. Although the means of many features are relatively close, the high T-statistics and extremely low p-values indicate statistically significant differences between the real and synthetic data. The Levene's test for equality of variances also highlights substantial discrepancies, with high Levene statistics and p-values near zero, suggesting significant variance differences between the two datasets. While a few features, such as P2_ATSW_Lamp and P2_MASW, have p-values above the typical significance threshold in the Levene test, the overall trend shows that the synthetic data deviates from the real data in both mean and variance for most features.

The t-statistics with p-values close to zero for most numeric features in the gas pipeline dataset indicate significant differences between real and synthetic means as presented in Table 17 of appendix A. Levene's test results for equality of variances also reveal significant differences, with several features showing large disparities in variance between real and synthetic data.

The statistical comparison between the numeric features of the real and synthetic Power System dataset reveals substantial differences across most features, as evidenced by the high T-statistics and extremely low p-values in Table 18 of appendix A. Furthermore, the Levene's test for equality of variances consistently shows large Levene statistics and near-zero p-values, confirming significant disparities in the variances of the two datasets. While a few features exhibit p-values that exceed typical significance thresholds for variance equality, such as certain voltage and current readings, the overall trend highlights that both the means and variances of synthetic data differ notably from the real data.

The discrepancies in means and variances between real and synthetic datasets in the HAI, Gas Pipeline, and Power System datasets indicate potential limitations in the synthetic data generation process. These differences may arise from DoppelGANger's challenges in accurately capturing complex temporal dependencies and feature correlations inherent in real CPS data. Additionally, underrepresented features in the real dataset may not be effectively learned by the GAN model, leading to inconsistencies between the synthetic and real data distributions.

4.3 Result comparison based on correlation analysis

A correlation threshold of 0.9 was set to identify highly correlated feature pairs Fig. 9 in Appendix B presents a visual comparison of the real and synthetic WDT datasets. Notably, no feature pairs in the WDT dataset exhibited correlations above the 0.9 threshold, indicating minimal multicollinearity concerns for this dataset. In the HAI dataset, there were 20 common feature pairs with correlations above 0.9, 44 highly correlated pairs unique to the real data, and 14 pairs only in the synthetic data. For the Gas Pipeline dataset, there were 12 feature pairs with common correlations above 0.9, no pairs of high correlation real data, and 8 highly correlated pairs found exclusively in the synthetic data. Similarly, in the Power System dataset, 113 common feature pairs were identified, with 32 pairs highly correlated only in the real data and 89 pairs found exclusively in the synthetic dataset. Figs. 10, 11, and 12 in Appendix B presents the correlation matrices of the HAI, gas pipeline and power system datasets.

The differences in correlation structures between real and synthetic CPS datasets can significantly impact the performance of ML models that rely on the relationships between features. Logistic regression assumes that features are independent, and high correlations between can lead to unreliable coefficient estimates and reduce the model's predictive accuracy. In the synthetic datasets, the presence of feature pairs with high correlations, especially if not present in the real data, may cause logistic regression models to overfit or misinterpret relationships, leading to distorted predictions. On the other hand, missing key correlations in synthetic data could cause the model to overlook important feature interactions that exist in the real data, resulting in underperformance.

4.4 Result comparison based on silhouette score

Table 10 presents the Silhouette scores comparisons for real and synthetic datasets across the four CPS domains: WDT, HAI, Gas Pipeline, and Power System. From the results, the synthetic datasets generally exhibit higher Silhouette scores to real datasets, indicating the DoppelGANger's failurel to capture subtle dependencies that exist in real CPS datasets, leading probably to over-simplified patterns.

4.5 Comparative analysis by PCA-Isolation forest

Considering the imbalanced nature of the real and synthetic data, accuracy alone might not be the best metric, and it's essential to examine the weighted precision and recall for a more comprehensive evaluation. Table 11 presents the results of accuracy together with the weighted precision, recall and precision of the various synthetic and real CPS datasets. While synthetic data generally achieves high performance metrics, the primary focus is on how closely these synthetic results mirror the real results. Among the datasets, the HAI exhibits the closest match between synthetic and real data performance. The synthetic dataset achieves a precision of 95%, a recall of 77%, an F1-score of 85%, and an accuracy of 77% compared to the real dataset, which achieves a precision of 93%, a recall of 77%, an F1-score of 84%, and an accuracy of 77%.

The hyperparameters indicating the configuration of the PCA-Isolation Forest model that performed best on the respective datasets during the hyperparameter tuning process by grid search technique are presented in Table 12. The proportion of anomalies in the real datasets varies, with a set proportion of 3% for the real portion of WDT, 1% for synthetic WDT, gas pipeline and power system and real power system, and 20% for the HAI and real gas pipeline datasets to optimize model performance. Conversely, the proportion of anomalies in all synthetic datasets was uniformly set to 1%. For real data, the max_features parameter, which represents the proportion of features considered for splitting at each

Table 10 Silhouette Score Comparison: Real vs. Synthetic CPS Data

Dataset	Silhouette score (real)	Silhouette score (synthetic)
WDT	0.1986	0.5901
HAI	0.1532	0.2240
Gas pipeline	0.5288	0.8589
Power system	0.3891	0.9143

Table 11 Comparative Assessment of PCA-Isolation Forest: Real vs. Synthetic CPS Data

Dataset	Precision (weighted)	Recall (weighted)	F1-score (weighted)	Accuracy
WDT_Synthetic Data	0.87	0.84	0.78	0.84
WDT_Real Data	0.68	0.80	0.73	0.80
HAI_Synthetic Data	0.95	0.77	0.85	0.77
HAI_Real Data	0.93	0.77	0.84	0.77
Gas-Pipeline_Synthetic	0.96	0.97	0.96	0.97
Gas-Pipeline_Real Data	0.94	0.78	0.85	0.78
Power-Sys_Synthetic	0.48	0.67	0.54	0.67
Power-Sys_Real	0.59	0.72	0.61	0.72

Table 12 PCA-Isolation Forest Hyperparameters

Dataset	Contamination	Max features	Max samples	N estimators
WDT_Synthetic	0.01	1.0	100	50
WDT_Real	0.03	0.8	100	100
HAI_Synthetic	0.2	0.8	100	100
HAI_Real	0.2	1.0	200	150
Gas-Pipeline_Synthetic	0.01	0.8	100	50
Gas-Pipeline_Real	0.2	1.0	100	150
Power-System_Real	0.01	1.0	100	150
Power-System_Synthetic	0.01	1.0	100	150

Table 13 Comparison of Quantile Stacked LSTM Model Performance

Dataset	Synthetic dataset	Real dataset
WDT_Training MAE	0.0112	0.0126
WDT_Validation MAE	0.0135	0.0103
HAI_Validation MAE	0.3497	9.0644e-04
HAI_Training MAE	0.3471	6.2245e-04
Gas-Pipeline_Training MAE	0.0291	0.0017
Gas-Pipeline_Validation MAE	0.0284	0.0021
Power-System_Training MAE	0.0194	0.0181
Power-System_Validation MAE	0.0196	0.0184

node, is set at 80% for the WDT and varies up to 100% for other datasets like HAI and gas pipeline, while for synthetic data, all the features are considered during the training process. The number of samples to draw from the datasets to train each base estimator and the number of base estimators (trees) in the ensemble are consistently set at 100 and 150, respectively, for real datasets in more complex environments such as HAI and Gas-Pipeline, ensuring robust model training and optimal results.

4.6 Comparative analysis by quantile stacked LSTM

The comparative analysis of the QS-LSTM model across multiple datasets is detailed in Table 13. Although the model performs significantly better on real data across all domains, demonstrating high efficacy in managing genuine operational conditions in water distribution, industrial control systems, gas pipelines, and power systems, the focus is on how closely the synthetic datasets replicate the performance of the real datasets. The power system dataset exhibits the closest results, with a training MAE of 0.0194 for the synthetic data compared to 0.0181 for the real data, and a validation MAE of 0.0196 versus 0.0184.

However, the consistently higher MAE on synthetic data, especially notable in the HAI and gas pipeline datasets, indicate a potential mismatch between the synthetic data's characteristics and the complexities of real-world scenarios.

Table 14 Comparison of Quantile Stacked LSTM Training Time

Dataset	Synthetic dataset	Real dataset
WDT_Training Time (s)	385	420
HAI_Training Time (s)	2005.17	2035.47
Gas-Pipeline_Training Time (s)	1543.90	1586.10
Power-System_Training Time (s)	5543.39	6176

Table 15 Comparison of training times for TimeGAN and DoppelGANger

Dataset	TimeGAN training time (s)	DoppelGANger training time (s)	TimeGAN (s/epoch)	DoppelGANger (s/epoch)
WDT	3069.85	365.25	30.70	3.65
Gas Pipeline	2961.45	840.95	56.94	8.41
Power System	3367.44	516.45	210.46	5.16
HAI	3325.91	1467.06	665.18	14.67

For TimeGAN, the training times are based on iterations, which are converted to approximate seconds per epoch(normalized to 100 epochs). DoppelGANger training times are reported directly for 100 epochs

Table 14 provides a comparison of training times for the QS-LSTM model across the different CPS datasets, differentiating between synthetic and real data. The training times offer insights into the computational efficiency and resource demands of the model in different scenarios. Across all datasets, the QS-LSTM model consistently requires less time to train on synthetic data compared to real data.

4.7 Computational efficiency: DoppelGANger vs TimeGAN

The implementation time for generating synthetic datasets using DoppelGANger is compared with that of TimeGAN and CTGAN to evaluate computational efficiency. The TimeGAN model was configured with a sequence length of 24, using a GRU-based architecture with hidden dimensions of 24 and three layers, and its training process involved 8,500 iterations per phase covering embedding network training, supervised loss training, and joint training for all CPS datasets. Table 15 compares the training performance of TimeGAN and DoppelGANger across several datasets by converting total training times into an average time per epoch. For TimeGAN, the conversion was done by estimating the number of epochs from the total number of generated sequences divided by the batch size and then normalizing the total training time to 100 epochs. For example, on the WDT dataset, 8,500 iterations roughly translate to 100 epochs, yielding an average of about 30.70 s per epoch. In contrast, DoppelGANger's training times are reported directly for 100 epochs, resulting in much lower per-epoch times. While these calculations provide useful insight into computational efficiency, it is important to note that the two models differ significantly in architecture. TimeGAN combines an autoencoder (embedding and recovery networks) with adversarial training through a generator, supervisor, and discriminator to capture temporal dynamics in time-series data, whereas DoppelGANger employs a distinct design that leverages both data augmentation and more streamlined adversarial training strategies. These architectural differences not only impact model performance but also affect training efficiency and resource utilization.

5 Discussion

This study addresses a critical challenge in the field of CPS being the lack of quality, available datasets required for the development, testing, and validation of models and algorithms. Collecting real-world CPS data is constrained by significant costs, time, and the expertise needed to manage and operate complex systems. In response, the paper proposes the use of DoppelGANger, a GAN based model, to generate synthetic data from real-world CPS datasets across various domains, including water distribution, industrial control systems, gas pipelines, and power systems. The synthetic data generated by DoppelGANger was validated through a comparative analysis with real-world data. Statistical evaluations, visualizations, correlation analysis, clustering performance and anomaly detection techniques were employed to assess the reliability and accuracy of the synthetic datasets.

The t-SNE visualizations provided a qualitative assessment of how well the synthetic datasets preserved the structure of real CPS data. The WDT dataset exhibited significant improvements in cluster alignment as training epochs increased, with the synthetic data closely resembling real-world patterns after 1000 epochs. However, for datasets such as HAI, Gas Pipeline, and Power System, the visual alignment remained inconsistent despite increased training iterations. The DoppelGANger model struggled to capture complex temporal dependencies in certain CPS environments. The divergence in t-SNE results, particularly in HAI and Power System datasets, highlights the need for further refinements in GAN architecture to improve the fidelity of synthetic time-series data.

A statistical analysis was conducted to compare mean values, variance, and overall distribution similarity between real and synthetic datasets. The t-test results showed that while some features, such as Tank_2 and Tank_8 in the WDT dataset, closely matched their real counterparts, others exhibited significant discrepancies. Particularly in the Gas Pipeline and Power System datasets, the synthetic data failed to replicate certain numerical distributions, suggesting that the GAN model struggled with preserving feature-specific statistical properties. Levene's test results indicated that variance mismatches were more pronounced in the HAI dataset.

The correlation analysis revealed notable differences in feature relationships between real and synthetic datasets. The WDT dataset exhibited minimal multicollinearity concerns, as the synthetic data did not introduce spurious high correlations. However, in the HAI dataset, 20 feature pairs maintained high correlations above 0.9 in both real and synthetic data, while 44 pairs were highly correlated only in real data, and 14 pairs were unique to the synthetic dataset. Similarly, in the Power System dataset, 113 common feature pairs were highly correlated, but 32 were exclusive to the real data, and 89 were synthetic-specific. These discrepancies in correlation structures can significantly impact ML models that rely on feature dependencies. For instance, logistic regression models assume independent features, and misaligned correlations in synthetic datasets could lead to distorted predictions or overfitting.

Silhouette score comparisons across datasets indicated that synthetic datasets generally exhibited higher silhouette values than real datasets. This was evident in the Power System dataset, where the synthetic data achieved a silhouette score of 0.9143 compared to 0.3891 for real data. While this suggests that the synthetic data formed well-defined clusters, it also raises concerns about oversimplification, where the synthetic data may not capture the true variability of real-world CPS. The Gas Pipeline dataset also showed a significant increase in silhouette score from 0.5288 (real) to 0.8589 (synthetic), indicating improved cluster compactness in the generated data. However, a higher silhouette score does not necessarily imply better data fidelity, as real CPS data often exhibits complex overlapping structures that are not easily separable into well-defined clusters.

To evaluate the anomaly detection performance of synthetic data, PCA-Isolation Forest was applied to both real and synthetic datasets. The weighted precision, recall, and F1-score results indicated that synthetic data achieved high precision in anomaly detection, with values of 87% for WDT, 95% for HAI, and 96% for Gas Pipeline. However, the performance dropped significantly for the Power System dataset, where precision fell to 48%, suggesting that the synthetic data in this domain did not accurately capture normal and anomalous patterns. The contamination parameter, which controls the proportion of anomalies in real datasets, was tuned to 1% for synthetic datasets across all domains. The model's accuracy varied, with the Gas Pipeline dataset performing best at 97%, while the Power System dataset remained a challenge, with an accuracy of only 67%. This discrepancy suggests that while synthetic data generation can enhance anomaly detection capabilities, domain-specific challenges persist.

While a DL anomaly detection model, QS-LSTM performed well on real datasets, it struggled with the synthetic datasets, especially in the case of the HAI and gas pipeline systems. This indicates that synthetic data may not always capture the subtle nuances required for high-performance anomaly detection. Further refinement in the generation process or a more domain-specific approach may be necessary to improve synthetic data's utility in these complex detection tasks.

5.1 Limitations and challenges

One of the limitations of this study is the need for high computational performance and advanced data management capabilities. This becomes critical when handling large volumes of synthetic data and complex anomaly detection models. It is essential to ensure that the computational resources and system architectures are adequately equipped to meet these demands.

Another limitation encountered was mode collapse, which led to the generator producing limited variations in synthetic data, thereby failing to capture the full diversity of real-world CPS datasets. This issue affected the distributional accuracy of the generated data, particularly in complex datasets such as HAI and Power System, where multiple

interdependent variables exist. As a result, certain patterns and rare events present in real data were not well-represented in the synthetic datasets.

Furthermore, acquiring real CPS datasets across different domains was challenging due to privacy concerns and security restrictions. Many CPS environments are proprietary, making publicly available datasets scarce and limited in scope. This constraint hinders benchmarking and validation of synthetic data models, emphasizing the need for alternative data-sharing strategies or domain-specific simulation frameworks.

Additionally, comparing computational efficiency between DoppelGANger and TimeGAN posed challenges due to their differing training architectures. While DoppelGANger follows an epoch-based training process, TimeGAN is trained using iterations, making direct comparisons of training time complex.

5.2 Practical applications of synthetic data in CPS

The generation of synthetic datasets using GAN-based models such as DoppelGANger and TimeGAN has significant implications for predictive maintenance, fault diagnosis, cybersecurity, and digital twins in CPS. These applications enable AI-driven solutions for complex industrial and infrastructure systems. Predictive maintenance relies on early detection of potential failures to minimize downtime and reduce operational costs. However, traditional ML models require large-scale labeled failure data, which is often scarce and imbalanced in real-world CPS. Synthetic data can help by training on quality synthetic sensor data, ensuring robustness across different system conditions. Zhang et al. generated smart grid dataset based on Pecan Street dataset using a general probabilistic time-series data model [48].

Supervisory Control and Data Acquisition (SCADA) systems are integral to managing industrial control processes in water treatment plants. However, training cybersecurity models such as intrusion detection systems, requires access to diverse attack datasets, which are often unavailable due to privacy concerns and security restrictions. To address this challenge, the authors of [53] utilized GAN-based models to generate synthetic network logs that simulate a range of cyberattack scenarios, including replay attacks, sensor spoofing, and malware-injected anomalies. Furthermore, TimeGAN was employed to create realistic SCADA sensor anomalies, replicating system behavior under simulated cyber intrusions. The integration of these synthetic attack datasets significantly enhanced IDS detection accuracy for previously unseen threats, enabling security teams to conduct comprehensive attack simulations without compromising real-world plant data. This approach not only strengthens cybersecurity preparedness but also ensures compliance with data privacy regulations, making it a viable solution for secure and scalable threat modeling.

Fault detection models often suffer from data imbalance, as real-world faults occur infrequently and collecting labeled fault data can be expensive. Generative models can augment training datasets with synthetic fault scenarios, improving model performance. Simulate transient and evolving failures to create more realistic failure datasets. and combine physics-based simulations with synthetic data to enhance fault realism. For example in manufacturing CPS, synthetic data can be used to train machine vision systems for defect detection in assembly lines, ensuring that AI models can recognize rare and complex faults without requiring massive amounts of real-world defect data [43].

6 Directions for future research

This study has highlighted the significant need for further research in synthetic dataset generation for CPS in enhancing realism and ensuring the fidelity of generated data. Future work could explore several improvements discussed in the subsequent paragraphs to refine and expand the applicability of generative models in this domain.

First, enhancing the ability of GAN models such as DoppelGANger and TimeGAN to effectively handle rare events is essential for CPS security applications. These models tend to underrepresent anomalies, which are critical for detecting cyberattacks, system failures, and rare operational deviations. To address this, anomaly-aware training strategies can be incorporated by implementing importance weighting in the training objective, ensuring that rare events receive greater emphasis during model optimization. Another promising approach involves leveraging self-supervised learning for rare event representation, where the model learns distinct feature embeddings for anomalies before GAN training. This technique enables the generator to better capture and synthesize rare event distributions, improving the overall diversity and usefulness of synthetic datasets.

Second, incorporating more advanced loss functions, such as those employed in WGANs, could help stabilize training and improve the fidelity of synthetic data. For example, the DoppelGANger discriminator loss function could be replaced with a focal loss function to focus on hard to classify samples to foster more diverse and realistic data generation.

Additionally, enhancing optimization techniques and incorporating gradient penalty methods may further mitigate mode collapse and improve convergence.

Third, exploring domain-specific GAN models tailored for CPS in different domains is a crucial direction for future research. Domain-adaptive generative models such as physics-informed GANs or transformer-based architectures instead of the LSTM in DoppelGANger, could better capture the unique characteristics of CPS. These models can leverage prior knowledge and system dynamics to generate synthetic data that is more representative of real-world operational behaviors.

Fourth, introducing post-processing techniques, such as feature-specific scaling corrections, time-series smoothing and distribution alignment methods could improve the usability of synthetic datasets for downstream ML tasks. Applying optimal transport methods or domain adaptation techniques to refine the synthetic data could bridge gaps between real and generated distributions.

Fifth, integrating federated learning into the DoppelGANger synthetic dataset generation process. Traditional GAN training relies on large, centralized datasets, which may not always be feasible due to data sensitivity, regulatory constraints, and storage limitations in CPS environments. Model training can therefore be decentralized across multiple edge devices, industrial facilities, or organizations, allowing data to remain localized while sharing only model updates instead of raw data.

Finally, extending the DoppelGANger architecture to support multimodal CPS datasets by integrating multi-stream generators to handle diverse data types, such as time-series sensor data, network traffic logs, and categorical system metadata, while incorporating modality-specific discriminators to improve training stability. A cross-modality attention mechanism can be introduced to capture dependencies between different modalities, ensuring more coherent synthetic data generation.

7 Conclusion

This study explored the generation of synthetic CPS datasets using the DoppelGANger model and evaluated their fidelity across multiple domains, including water distribution, industrial control, gas pipelines, and power systems. The synthetic datasets were assessed through statistical comparisons, clustering metrics, and anomaly detection models. The findings suggest that while synthetic CPS data generated by DoppelGANger presents a promising solution to the scarcity of real-world CPS datasets, substantial further research is needed to fully explore and validate its potential.

The t-SNE visualizations revealed that the synthetic data improved in structural alignment with real data as training epochs increased in the WDT dataset. However, HAI, Gas pipeline and Power System datasets showed inconsistencies. Additionally, the study emphasizes caution in anomaly detection applications: while DL models such as QS-LSTM perform robustly on real data, synthetic datasets in more complex domains like HAI and gas pipelines struggle to capture the nuanced patterns necessary for effective detection. The reduced training time makes DoppelGANger a viable option for large-scale CPS synthetic dataset generation in time-sensitive applications.

Overall, synthetic data presents significant opportunity to overcome barriers in CPS data scarcity research such as cost, time, and expertise, while enabling the simulation of a wide range of scenarios. These findings encourage further development of advanced synthetic data generation techniques and their integration into CPS design, optimization, and control, particularly in critical domains such as water distribution, energy management, and autonomous systems. The study emphasizes the need for advanced synthetic data generation techniques for CPS that can better mimic real-world complexities and support robust adaptable models.

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Author contributions YA wrote the main manuscript text, all authors reviewed the manuscript. KR made substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data; or the creation of new software used in the work; KR drafted the work or revised it critically for important intellectual content; KR, YA approved the version to be published; and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability The data used for this experiment are available in public repositories. The detailed information is given in the methodology section.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no Conflict of interest.

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Appendix A: Statistical Analysis for the HAI, Gas pipeline and Power systems CPS Datasets

See Tables 16, 17 and 18

Table 16 Statistical Comparison of Real and Synthetic Data for HAI Dataset

Feature	Mean (Real)	Mean (Synthetic)	Std Dev (Real)	Std Dev (Synthetic)	T-Statistic	P-value (T-Test)	Levene-Statistic	P-value (Levene Test)
P1_B2004	0.524	0.510	0.317	0.371	14.090	0.000	34306.466	0.000
P1_B2016	0.410	0.425	0.135	0.144	-35.653	0.000	1290.420	0.000
P1_B3004	0.528	0.571	0.357	0.336	-42.130	0.000	4368.240	0.000
P1_B3005	0.356	0.283	0.309	0.254	87.616	0.000	9191.557	0.000
P1_B4002	0.370	0.361	0.470	0.462	6.079	0.000	861.514	0.000
P1_B4005	0.129	0.119	0.171	0.165	20.262	0.000	3497.354	0.000
P1_B400B	0.130	0.087	0.171	0.105	102.979	0.000	28887.960	0.000
P1_B4022	0.392	0.396	0.242	0.259	-4.955	0.000	3119.574	0.000
P1_FCV01D	0.173	0.149	0.138	0.147	56.905	0.000	1989.260	0.000
P1_FCV01Z	0.162	0.124	0.140	0.148	90.773	0.000	4888.258	0.000
P1_FCV02D	0.741	0.793	0.427	0.351	-45.340	0.000	23322.802	0.000
P1_FCV02Z	0.699	0.759	0.361	0.314	-60.132	0.000	34473.592	0.000
P1_FCV03D	0.390	0.366	0.092	0.105	82.648	0.000	1020.219	0.000
P1_FCV03Z	0.398	0.410	0.102	0.122	-34.552	0.000	6714.295	0.000
P1_FT01	0.146	0.147	0.029	0.051	-4.774	0.000	109681.594	0.000
P1_FT01Z	0.246	0.239	0.039	0.072	43.014	0.000	128540.970	0.000
P1_FT02	0.053	0.034	0.134	0.074	59.176	0.000	5435.936	0.000
P1_FT02Z	0.130	0.112	0.172	0.165	34.692	0.000	2609.845	0.000
P1_FT03	0.560	0.526	0.139	0.121	88.428	0.000	2382.847	0.000
P1_FT03Z	0.609	0.592	0.137	0.131	42.628	0.000	162.161	0.000
P1_LCV01D	0.160	0.146	0.039	0.056	96.198	0.000	28978.980	0.000
P1_LCV01Z	0.153	0.159	0.038	0.096	-23.736	0.000	35027.666	0.000
P1_LIT01	0.516	0.519	0.071	0.110	-10.967	0.000	29952.987	0.000
P1_PCV01D	0.403	0.428	0.049	0.101	-107.729	0.000	109919.299	0.000

Table 16 (continued)

Feature	Mean (Real)	Mean (Synthetic)	Std Dev (Real)	Std Dev (Synthetic)	T-Statistic	P-value (T-Test)	Levene-Statistic	P-value (Levene Test)
P1_PCV01Z	0.399	0.402	0.049	0.127	− 9.275	0.000	81553.669	0.000
P1_PCV02Z	0.018	0.016	0.020	0.020	33.456	0.000	4691.926	0.000
P1_PIT01	0.548	0.557	0.056	0.081	− 43.013	0.000	21581.658	0.000
P1_PIT02	0.162	0.143	0.233	0.179	31.462	0.000	3309.777	0.000
P1_PP04	0.583	0.613	0.356	0.340	− 29.841	0.000	1798.116	0.000
P1_PP04SP	0.718	0.728	0.404	0.367	− 8.572	0.000	3972.720	0.000
P1_TIT01	0.467	0.450	0.199	0.197	30.078	0.000	1313.132	0.000
P1_TIT02	0.517	0.500	0.275	0.260	21.073	0.000	14.988	0.000
P1_TIT03	0.454	0.459	0.209	0.240	− 6.745	0.000	7132.533	0.000
P2_24Vdc	0.492	0.571	0.117	0.109	− 238.984	0.000	7799.416	0.000
P2_ATSW_Lamp	0.970	0.959	0.171	0.172	21.038	0.000	3.653	0.056
P2_AutoGO	0.970	0.957	0.171	0.172	25.741	0.000	16.696	0.000
P2_AutoSD	0.466	0.520	0.267	0.246	− 71.387	0.000	1215.631	0.000
P2_MASW	0.030	0.035	0.171	0.170	− 9.348	0.000	0.624	0.430
P2_MASW_Lamp	0.030	0.035	0.171	0.170	− 9.393	0.000	0.569	0.451
P2_ManualGO	0.030	0.035	0.171	0.170	− 9.402	0.000	0.475	0.491
P2_ManualSD	0.183	0.111	0.318	0.225	89.108	0.000	18036.818	0.000
P2_SCO	0.559	0.561	0.093	0.136	− 5.195	0.000	26243.964	0.000
P2_SCST	0.533	0.585	0.123	0.139	− 136.931	0.000	754.579	0.000
P2_SIT01	0.569	0.562	0.105	0.116	19.992	0.000	4224.268	0.000
P2_VIBTR01	0.534	0.624	0.210	0.187	− 153.444	0.000	10065.187	0.000
P2_VIBTR02	0.516	0.512	0.213	0.252	4.623	0.000	8828.333	0.000
P2_VIBTR03	0.519	0.506	0.176	0.165	26.868	0.000	2023.142	0.000
P2_VIBTR04	0.502	0.486	0.228	0.266	21.619	0.000	17264.830	0.000
P2_VT01	0.537	0.585	0.234	0.248	− 68.603	0.000	80.314	0.000
P3_FIT01	0.161	0.172	0.245	0.247	− 15.734	0.000	274.771	0.000
P3_LCP01D	0.277	0.222	0.288	0.226	72.805	0.000	29093.393	0.000
P3_LCV01D	0.344	0.419	0.326	0.323	− 78.852	0.000	2172.318	0.000
P3_LIT01	0.439	0.424	0.246	0.232	21.909	0.000	1438.672	0.000
P3_PIT01	0.128	0.072	0.217	0.115	111.507	0.000	73561.810	0.000
P4_HT_FD	0.468	0.444	0.036	0.133	84.654	0.000	356689.644	0.000
P4_HT_PO	0.380	0.437	0.364	0.355	− 53.795	0.000	3902.900	0.000
P4_HT_PS	0.384	0.389	0.390	0.412	− 4.303	0.000	9633.685	0.000
P4_LD	0.505	0.534	0.222	0.267	− 41.231	0.000	24699.582	0.000
P4_ST_FD	0.534	0.539	0.070	0.129	− 13.849	0.000	21535.793	0.000
P4_ST_GOV	0.393	0.376	0.138	0.174	36.915	0.000	30963.238	0.000
P4_ST_LD	0.391	0.424	0.146	0.215	− 61.741	0.000	60835.170	0.000
P4_ST_PO	0.393	0.390	0.138	0.144	8.696	0.000	2370.841	0.000
P4_ST_PS	0.750	0.754	0.307	0.310	− 4.012	0.000	77.876	0.000
P4_ST_PT01	0.533	0.551	0.056	0.089	− 82.410	0.000	30149.444	0.000
P4_ST_TT01	0.467	0.477	0.195	0.181	− 17.635	0.000	1742.127	0.000

Table 17 Statistical Comparison of Real and Synthetic Data for Gas Pipeline Dataset

Features	Mean (Real)	Mean (Synthetic)	Std Dev (Real)	Std Dev (Synthetic)	T-Statistic	P-value (T-Test)	Levene Statistic	P-value (Levene Test)
CRC	0.882	0.807	0.129	0.123	60.928	0.0	73.827	0.000
DataLength	0.081	0.048	0.096	0.077	38.809	0.0	1276.419	0.000
InvalidDataLength	0.006	− 0.014	0.079	0.073	26.462	0.0	5.518	0.019
PIDCycleTime	0.009	− 0.019	0.093	0.023	42.511	0.0	200.810	0.000
PIDDeadband	0.009	− 0.019	0.093	0.023	42.471	0.0	200.695	0.000
PIDGain	0.009	− 0.019	0.093	0.023	42.470	0.0	200.783	0.000
PIDReset	0.009	− 0.019	0.093	0.023	42.505	0.0	200.928	0.000
PipelinePSI	0.998	1.010	0.043	0.015	− 37.925	0.0	33.024	0.000
SetPoint	0.003	− 0.007	0.038	0.088	15.517	0.0	420.394	0.000
TimeInterval	0.022	− 0.006	0.012	0.031	119.478	0.0	12392.576	0.000
deltaPIDCycleTime	0.501	0.518	0.059	0.023	− 38.874	0.0	30.954	0.000
deltaPIDDeadband	0.501	0.504	0.059	0.037	− 5.852	0.0	71.987	0.000
deltaPIDGain	0.501	0.496	0.059	0.034	10.714	0.0	62.490	0.000
deltaPIDRate	0.000	− 0.019	0.000	0.023	122.178	0.0	2958.794	0.000
deltaPIDReset	0.501	0.494	0.059	0.051	13.371	0.0	1806.516	0.000
deltaPipelinePSI	0.500	0.483	0.006	0.031	77.250	0.0	2792.141	0.000
deltaSetPoint	0.500	0.464	0.012	0.027	174.659	0.0	764.190	0.000

Table 18 Statistical Comparison of Real and Synthetic Data for Power System Dataset

Feature	Mean (Real)	Mean (Synthetic)	Std Dev (Real)	Std Dev (Synthetic)	T-Statistic	P-value (T-Test)	Levene-statistic	P-value (Levene Test)
R1-PA1:VH	0.456	0.761	0.280	0.035	− 289.086	0.0	150093.923	0.0
R1-PM1:V	0.886	0.961	0.049	0.051	− 275.251	0.0	17445.956	0.0
R1-PA2:VH	0.506	0.957	0.311	0.051	− 384.852	0.0	183376.476	0.0
R1-PM2:V	0.893	0.968	0.015	0.052	− 360.224	0.0	65071.760	0.0
R1-PA3:VH	0.519	0.137	0.270	0.029	377.013	0.0	140667.013	0.0
R1-PM3:V	0.890	0.967	0.016	0.052	− 368.048	0.0	64257.297	0.0
R1-PA4:IH	0.458	0.702	0.281	0.028	− 231.819	0.0	162759.901	0.0
R1-PM4:I	0.220	0.067	0.079	0.037	468.988	0.0	2335.927	0.0
R1-PA5:IH	0.509	0.836	0.310	0.039	− 280.713	0.0	202025.354	0.0
R1-PM5:I	0.326	− 0.050	0.072	0.047	1152.723	0.0	9533.239	0.0
R1-PA6:IH	0.519	0.925	0.270	0.048	− 397.785	0.0	121154.040	0.0
R1-PM6:I	0.310	0.020	0.072	0.040	931.423	0.0	16925.237	0.0
R1-PA7:VH	0.456	0.910	0.280	0.046	− 428.311	0.0	132773.271	0.0
R1-PM7:V	0.892	0.968	0.025	0.052	− 342.666	0.0	46486.227	0.0
R1-PA8:VH	0.499	0.361	0.033	0.017	983.471	0.0	3122.475	0.0
R1-PM8:V	0.006	− 0.071	0.058	0.049	267.715	0.0	23275.152	0.0
R1-PA9:VH	0.501	0.355	0.031	0.018	1081.847	0.0	5038.603	0.0
R1-PM9:V	0.004	− 0.070	0.050	0.049	282.095	0.0	35694.337	0.0
R1-PA10:IH	0.458	0.873	0.281	0.043	− 391.177	0.0	138810.434	0.0
R1-PM10:I	0.318	0.874	0.076	0.043	− 1697.637	0.0	20479.953	0.0
R1-PA11:IH	0.496	0.705	0.193	0.029	− 286.702	0.0	29493.938	0.0
R1-PM11:I	0.017	− 0.071	0.076	0.049	256.859	0.0	7843.397	0.0
R1-PA12:IH	0.517	0.074	0.207	0.037	564.535	0.0	30388.789	0.0

Table 18 (continued)

Feature	Mean (Real)	Mean (Synthetic)	Std Dev (Real)	Std Dev (Synthetic)	T-Statistic	P-value (T-Test)	Levene-statistic	P-value (Levene Test)
R1-PM12:I	0.013	− 0.071	0.063	0.049	278.033	0.0	16204.403	0.0
R1:F	0.511	0.820	0.011	0.038	− 1994.421	0.0	59668.956	0.0
R1:DF	0.604	0.961	0.012	0.051	− 1750.986	0.0	86597.525	0.0
R1-PA:Z	0.008	− 0.063	0.009	0.048	375.759	0.0	311221.990	0.0
R1-PA:ZH	0.504	0.956	0.032	0.051	− 1966.016	0.0	28314.772	0.0
R1:S	0.002	− 0.071	0.048	0.049	281.099	0.0	45500.538	0.0
R2-PA1:VH	0.458	0.248	0.283	0.020	198.713	0.0	171330.535	0.0
R2-PM1:V	0.896	0.965	0.072	0.051	− 207.419	0.0	6417.105	0.0
R2-PA2:VH	0.511	0.954	0.310	0.050	− 377.745	0.0	180284.066	0.0
R2-PM2:V	0.903	0.967	0.019	0.052	− 303.741	0.0	52682.929	0.0
R2-PA3:VH	0.517	0.593	0.269	0.022	− 75.991	0.0	159873.200	0.0
R2-PM3:V	0.829	0.959	0.016	0.051	− 634.443	0.0	58005.159	0.0
R2-PA4:IH	0.547	0.711	0.283	0.029	− 154.989	0.0	168928.449	0.0
R2-PM4:I	0.276	0.612	0.082	0.023	− 1052.345	0.0	33463.742	0.0
R2-PA5:IH	0.481	0.309	0.268	0.018	171.872	0.0	153288.081	0.0
R2-PM5:I	0.326	0.489	0.072	0.016	− 589.686	0.0	127049.043	0.0
R2-PA6:IH	0.480	0.587	0.307	0.021	− 93.577	0.0	234844.244	0.0
R2-PM6:I	0.312	0.063	0.072	0.036	818.811	0.0	24348.546	0.0
R2-PA7:VH	0.458	0.709	0.283	0.028	− 236.948	0.0	167155.812	0.0
R2-PM7:V	0.898	0.967	0.032	0.052	− 297.866	0.0	31620.853	0.0
R2-PA8:VH	0.501	0.542	0.033	0.018	− 288.631	0.0	239.810	0.0
R2-PM8:V	0.009	− 0.071	0.088	0.049	212.815	0.0	6770.610	0.0
R2-PA9:VH	0.500	0.575	0.031	0.019	− 540.189	0.0	567.301	0.0
R2-PM9:V	0.006	− 0.071	0.063	0.049	256.345	0.0	19120.765	0.0
R2-PA10:IH	0.547	0.030	0.283	0.042	485.389	0.0	141231.638	0.0
R2-PM10:I	0.320	− 0.038	0.072	0.046	1109.467	0.0	10214.229	0.0
R3-PA7:VH	0.458	0.646	0.283	0.026	− 177.820	0.0	16846.580	0.0
R3-PM7:V	0.900	0.966	0.032	0.052	− 283.245	0.0	31586.208	0.0
R3-PA8:VH	0.501	0.333	0.033	0.016	1203.998	0.0	3253.372	0.0
R3-PM8:V	0.009	− 0.068	0.088	0.049	204.757	0.0	6602.706	0.0

Appendix B: Correlation Analysis Comparison for Real and Synthetic CPS Datasets

See Figures 9, 10, 11 and 12

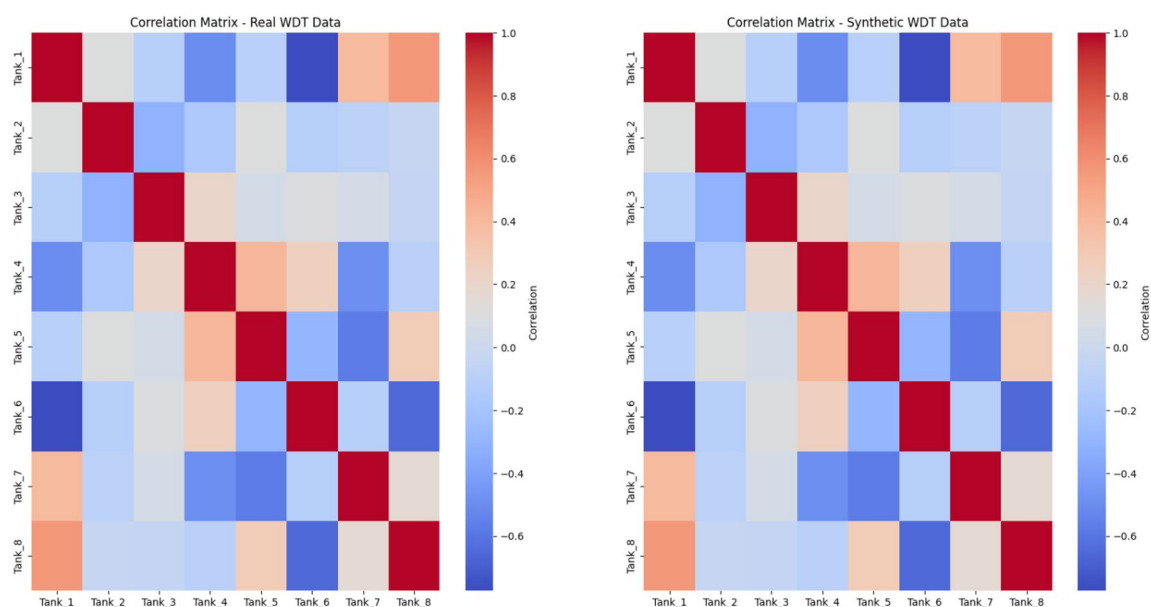


Fig. 9 Correlation Matrices for Real and Synthetic WDT Dataset

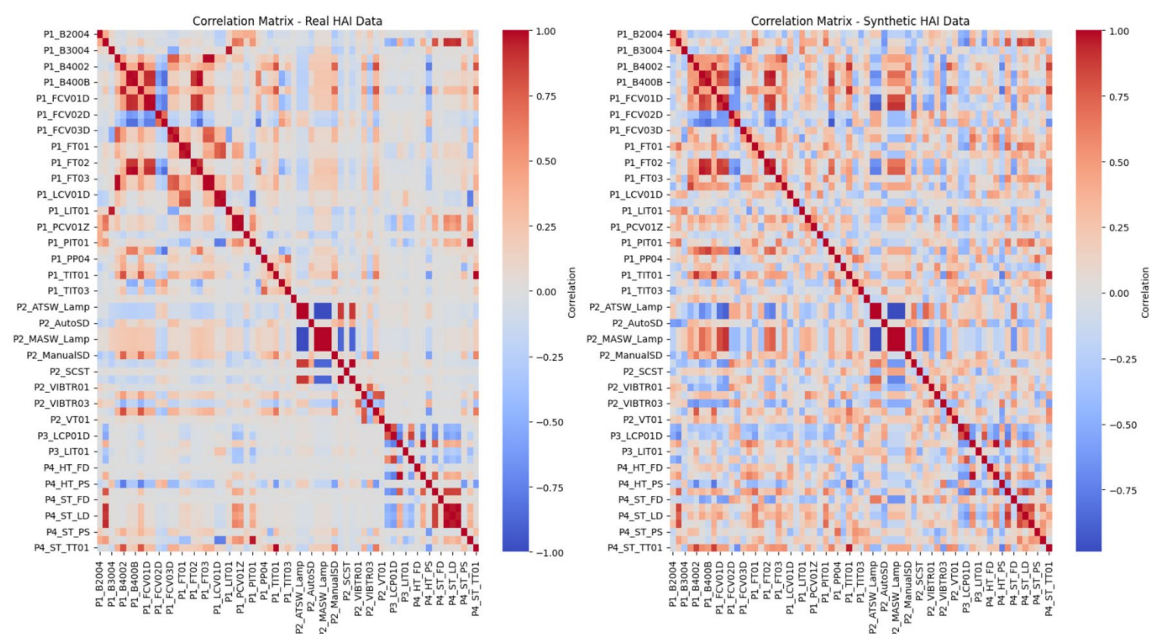


Fig. 10 Correlation Matrices for Real and Synthetic HAI Dataset

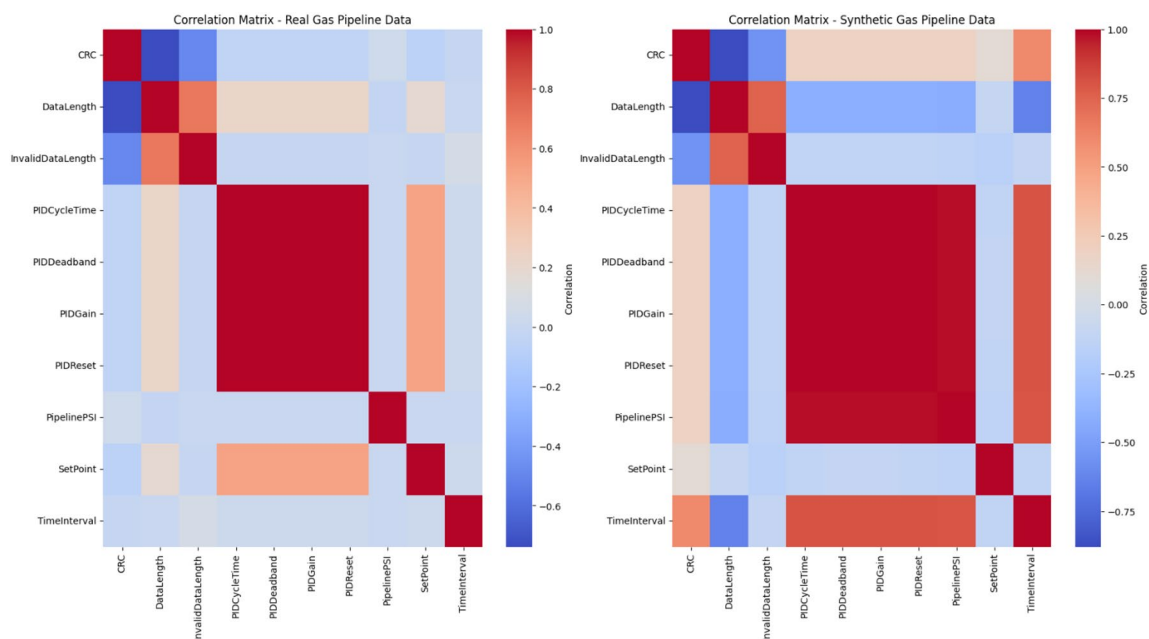


Fig. 11 Correlation Matrices for Real and Synthetic Gas Pipeline Dataset

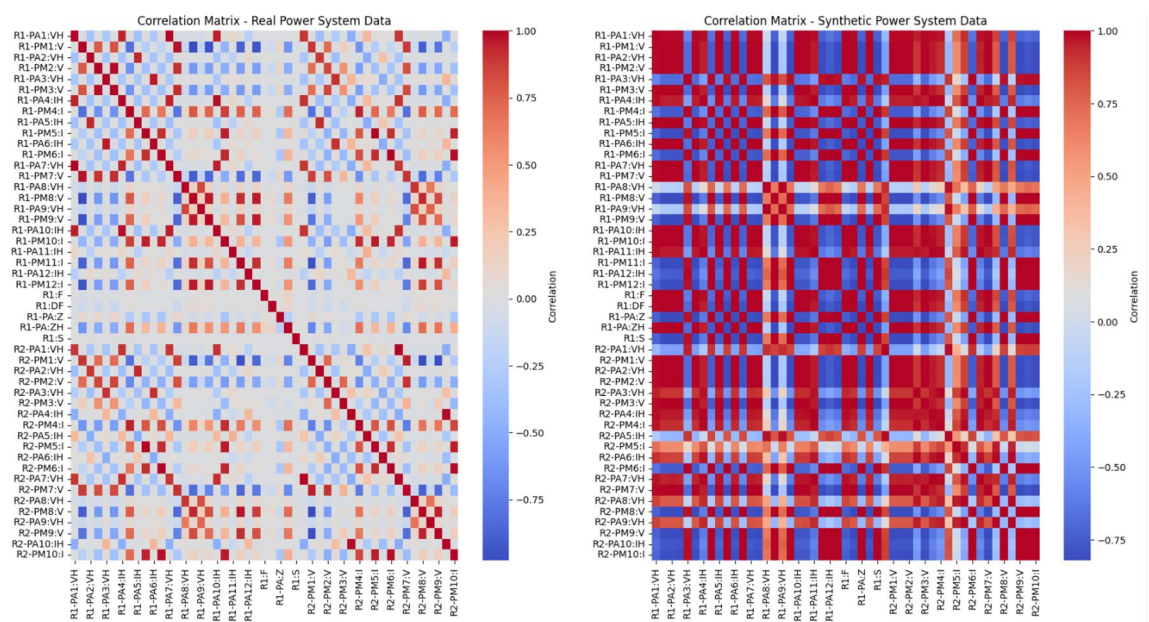


Fig. 12 Correlation Matrices for Real and Synthetic Power System Dataset

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